Neurocomputing 709 (2027) 135289

Contents lists available at ScienceDirect

Neurocomputing

journal homepage: www.elsevier.com/locate/neucom

CityGrounding: Graph-structured relation modeling for urban-scale 3D visual grounding

Xiaorong Zhang[a], Lingfei Chen[a], Yang Luo[b], Ting Han[b], Zilong Huang[b], Guorong Cai[a], Yiping Chen[b], Jinhe Su[a,*]

[a] The College of Computer Engineering, Jimei University, Xiamen, 361027, Fujian Province, China
[b] The School of Geospatial Engineering and Science, Sun Yat-sen University, Zhuhai, 519082, Guangdong Province, China

HIGHLIGHTS

- CityRel-Net enables relation-aware grounding in urban point clouds.
- GSREM parses language descriptions into semantic graphs for structured grounding.
- NGRE exploits neighboring instances to support robust contextual reasoning.
- A high-density real-world urban grounding dataset is introduced.
- CityRel-Net achieves state-of-the-art results on three benchmarks.

ARTICLE INFO

Communicated by B. Fan

Keywords:
3D visual grounding
Urban point clouds
Relation-aware reasoning
Cross-modal alignment

ABSTRACT

3D visual grounding aims to localize target objects in point clouds according to natural language descriptions. Although recent approaches have achieved promising results in indoor environments, grounding in outdoor urban scenes remains challenging because of complex linguistic relations and the visual homogeneity of densely distributed objects. Existing methods largely rely on implicit sequence modeling and isolated instance representations, which are insufficient for capturing fine-grained linguistic dependencies and contextual spatial relationships. To address these limitations, we propose CityRel-Net, a novel framework for urban-scale 3D visual grounding that explicitly models cross-modal relational structures. The proposed framework consists of two key modules: (1) the **Graph-Structured Relation Extraction Module** parses natural language descriptions into semantic graphs, enabling explicit modeling of grammatical structures and spatial relations; and (2) the **Neighbor-Guided Relation Enhancement** module strengthens relational reasoning by aggregating contextual information from spatially neighboring instances, thereby improving discrimination in visually cluttered urban environments. We additionally construct a real-world urban benchmark to evaluate the proposed method under dense and complex outdoor conditions. The dataset contains 32,901 descriptions and 11,000 object instances across diverse urban categories. Extensive experiments demonstrate that CityRel-Net achieves consistent state-of-the-art performance across multiple benchmarks, improving grounding accuracy from 27.05% to 58.29% (+**31.24%**) on CityRefer, 47.81% to 51.68% (+**3.87%**) on CityAnchor, and 67.90% to 69.83% (+**1.93%**) on the proposed dataset.

1. Introduction

3D visual grounding is a fundamental vision-language task that aims to localize target objects in 3D point cloud scenes from natural language descriptions [23], typically in the form of bounding boxes [13,14,27,31,41,42] or segmentation masks [44,45]. As a representative problem in multimodal learning, it bridges linguistic semantics and 3D spatial perception, and provides an important foundation for a variety of downstream applications such as autonomous driving [12,21,24,30,48], urban-scene understanding [22,29], and human–robot interaction

* Corresponding author.
Email addresses: zhangxr@jmu.edu.cn (X. Zhang), 202511835004@jmu.edu.cn (L. Chen), luoy583@mail2.sysu.edu.cn (Y. Luo), hant23@mail2.sysu.edu.cn (T. Han), huangzlong5@mail2.sysu.edu.cn (Z. Huang), guorongcai@jmu.edu.cn (G. Cai), chenyp79@mail.sysu.edu.cn (Y. Chen), sujh@jmu.edu.cn (J. Su).

https://doi.org/10.1016/j.neucom.2026.135289
Received 25 March 2026; Received in revised form 15 September 2026; Accepted 2 October 2026
Available online 6 October 2026
0925-2312/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

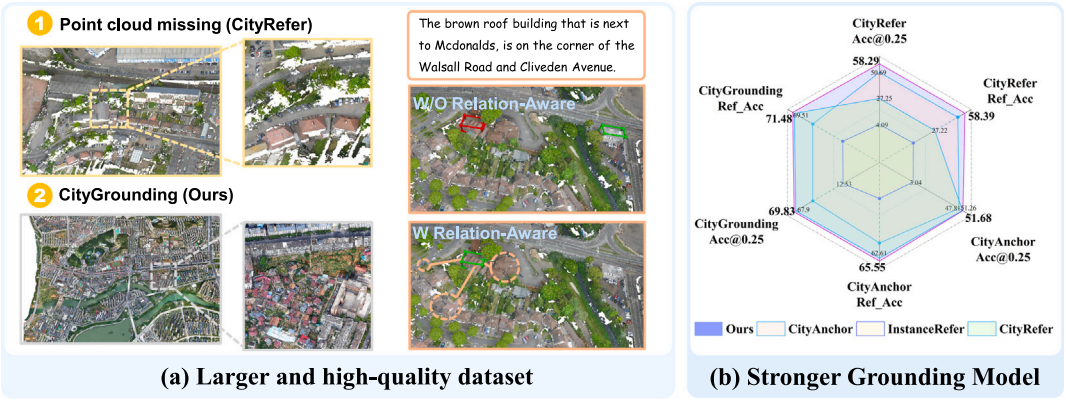


Fig. 1. (a) Previous datasets suffer from incomplete point clouds and limited scene coverage. We construct a **larger-scale and higher-quality** urban visual grounding dataset using UAV-captured city scenes. The right side shows results before and after applying the relation-aware module, where light Orange dashed circles mark the neighbors of the target instance. (b) our method achieves **state-of-the-art performance** on the CityRefer, CityAnchor, and CityGrounding datasets, where Ref_Acc denotes the proportion of correctly grounded target instances. The dashed lines indicate that the corresponding values are not available.

[20,38]. Although considerable progress has been made in recent years, most existing studies focus primarily on indoor environments, where scenes are relatively limited in spatial scale, object density, and semantic complexity. By contrast, outdoor urban scenes are substantially more challenging due to their larger spatial extent, more diverse object categories, and more complex linguistic expressions. These characteristics make urban 3D visual grounding significantly more difficult and expose the limitations of current grounding approaches (Fig. 1).

Existing approaches for urban visual grounding still suffer from several notable shortcomings. CityRefer [29], for example, extends the InstanceRefer [47] framework from indoor to outdoor environments with only limited architectural adaptation, leaving its language modeling and inter-object reasoning capacities largely underdeveloped. CityAnchor [22] improves grounding performance through a two-stage pipeline that combines coarse grounding with fine-grained matching based on multimodal large language models (MLLMs). However, its reliance on data quality, manual region-of-interest adjustment, and task-specific fine-tuning restricts its scalability and practical applicability across diverse urban scenes. More fundamentally, current methods often struggle to model the rich relational dependencies embedded in urban language descriptions. Although several indoor grounding methods [43,45] have explored relation-aware reasoning, such strategies remain insufficient for large-scale city environments characterized by dense object layouts, visual homogeneity, and diverse spatial interactions. Most prior relation-aware grounding methods mainly focus on indoor scenes, where object layouts are relatively compact and the surrounding context is less diverse. In contrast, urban point clouds contain many visually similar instances and long-range spatial dependencies, requiring relation modeling from both linguistic structures and neighboring urban entities. Moreover, most text encoders still process language as flat token sequences, overlooking grammatical structure and spatial clauses. As a result, they have limited ability to reason over complex expressions commonly found in urban scenarios, such as “the bus stop behind the red car near the mall,” where accurate grounding depends on understanding hierarchical and compositional relations.

To overcome these limitations, we propose **CityRel-Net**, a novel framework for urban-scale 3D visual grounding that explicitly models relational structures in both language and scene representations. Specifically, we introduce the **Neighbor-Guided Relation Enhancement** (**NGRE**) module, which aggregates spatial and semantic cues from neighboring objects to construct context-aware instance representations, thereby improving target discrimination in dense urban scenes. In addition, we develop the **Graph-Structured Relation Extraction Module** (**GSREM**), which represents textual descriptions as dependency graphs and models fine-grained grammatical and semantic dependencies through edge-aware attention. Different from previous relation-aware methods that mainly rely on object-level context or generic attention, our framework jointly exploits dependency-aware language structures and instance-level neighbor relations, making it more suitable for complex urban grounding scenarios. By jointly enhancing contextual object reasoning and structured language understanding, the proposed framework enables more robust and precise grounding in complex urban environments.

Secondly, the limitations of existing urban-scene visual grounding datasets further constrain current research. CityRefer [29] provides 35k descriptions based on SensatUrban [17], yet it covers only a limited range of object categories and suffers from incomplete point clouds with blank regions. CityAnchor [22] expands category coverage through synthetic data from STPLS3D [4], but the dataset contains only 1448 textual expressions. Moreover, recent datasets leveraging LLM techniques, such as Mono3DVG [48] and CityQA [36], primarily focus on road scenes or medium-scale point clouds, which limit their generalizability to complex real-world urban environments. To address these limitations, we construct a new large-scale, high-density, centimeter-level real-world urban visual grounding dataset with the assistance of a multimodal large language model. Considering that MLLM-assisted annotation may introduce noise, such as ambiguous descriptions, incorrect spatial relations, or target-description mismatch, we further employ structured prompts, instance-level filtering, automatic invalid-output removal, and manual sampling-based validation to improve annotation reliability.

In summary, the main contributions of this paper are as follows:

- We propose **CityRel-Net**, a novel framework for urban-scale 3D visual grounding that explicitly models cross-modal relational structures, addressing the limitations of implicit sequence modeling and isolated instance reasoning in dense urban scenes.
- We develop two complementary modules, namely the **Neighbor-Guided Relation Enhancement** (**NGRE**) module and the **Graph-Structured Relation Extraction Module** (**GSREM**), to capture contextual object interactions and fine-grained linguistic dependencies for relation-aware grounding.
- We also construct CityGrounding as an additional real-world benchmark for dense urban scenes. Extensive experiments on CityRefer, CityAnchor, and the proposed dataset demonstrate that CityRel-Net achieves state-of-the-art performance and generalization across diverse urban grounding scenarios.

2. Related work

2.1. 3D visual grounding

As summarized in the recent survey [11], visual grounding has evolved from 2D image-region localization to 3D object localization in point clouds, requiring models to align language not only with visual appearance but also with geometric structures and spatial layouts. The task of 3D visual grounding aims to localize target objects in 3D scenes given natural language descriptions [23]. Existing methods can be broadly categorized into two-stage and one-stage approaches. Two-stage methods [9,18,34,35,50] first generate object proposals using pre-trained 3D detectors or segmentation heads [5,25,39], and then match them with textual features encoded by models such as BERT [7]. Feature alignment is typically achieved via Transformer-based attention [49] or contrastive learning [1]. Representative two-stage methods include ScanRefer [3] and ReferIt3D [2], which utilize pre-trained 3D object detectors to obtain proposals that semantically match textual descriptions. In contrast, one-stage methods [14,33,40,46] integrate detection and grounding into a unified pipeline, directly regressing the target bounding box by globally matching point-wise features and textual embeddings. To further enrich contextual representations, ViewRefer [15] introduces multi-view knowledge into 3D visual grounding, showing the importance of additional context for language-guided localization. Unlike ViewRefer, our work focuses on relation-aware context modeling in urban point clouds. Despite significant progress, most studies remain constrained to indoor environments.

Directly transferring indoor grounding methods to large-scale outdoor scenes suffers from performance degradation due to memory constraints and feature sparsity caused by point cloud downsampling.

2.2. Outdoor 3D visual grounding benchmarks

Compared with indoor point clouds, urban-scale point clouds are substantially larger and more complex, involving diverse object categories such as vehicles, buildings, trees, and street furniture. Most existing outdoor datasets, including Semantic3D [16], Toronto-3D [37], and SensatUrban [17], mainly focus on segmentation or classification tasks and lack the language annotations required for visual grounding. Recently, CityRefer [29] and CityAnchor [22] have extended this line of research by introducing urban-scale grounding benchmarks that support cross-modal alignment between natural language and 3D scenes. Nevertheless, these datasets still suffer from limitations in point cloud completeness, dataset scale, or real-world fidelity, which restrict their ability to serve as comprehensive benchmarks for realistic urban grounding. To address these issues, we introduce a new high-density real-world urban visual grounding dataset.

3. Dataset

To address the limitations of existing 3D visual grounding datasets in outdoor environments, we construct CityGrounding, the first centimeter-level, real-world urban-scale dataset automatically annotated using multimodal large language models (MLLMs). Unlike previous approaches that utilize MLLMs for dataset construction, our method integrates instance-level segmentation masks and relational scene graphs to ensure a precise correspondence between textual descriptions and 3D instances within the scene. The source point clouds cover a combined area of 25 km^2 across two cities. From these source data, we select 459 non-overlapping blocks, each measuring 100 m × 100 m. These selected blocks form the grounding scenes in CityGrounding and have a total ground footprint of 4.59 km^2. The final benchmark contains 32,901 descriptions associated with 11,000 object instances.

3.1. Dataset construction

As shown in Fig. 2, the annotation pipeline consists of two stages: (i) acquisition of urban-scale Instance Segmentation masks and (ii) automatic generation of natural language descriptions. In the first stage, we employ a SoftGroup-based segmentation network pre-trained on STPLS3D to obtain fine-grained 3D instance masks.

To ensure the accuracy and semantic diversity of the segmented instances, we apply an IoU-based filtering strategy together with manual sampling-based validation, resulting in reliable segmentation across

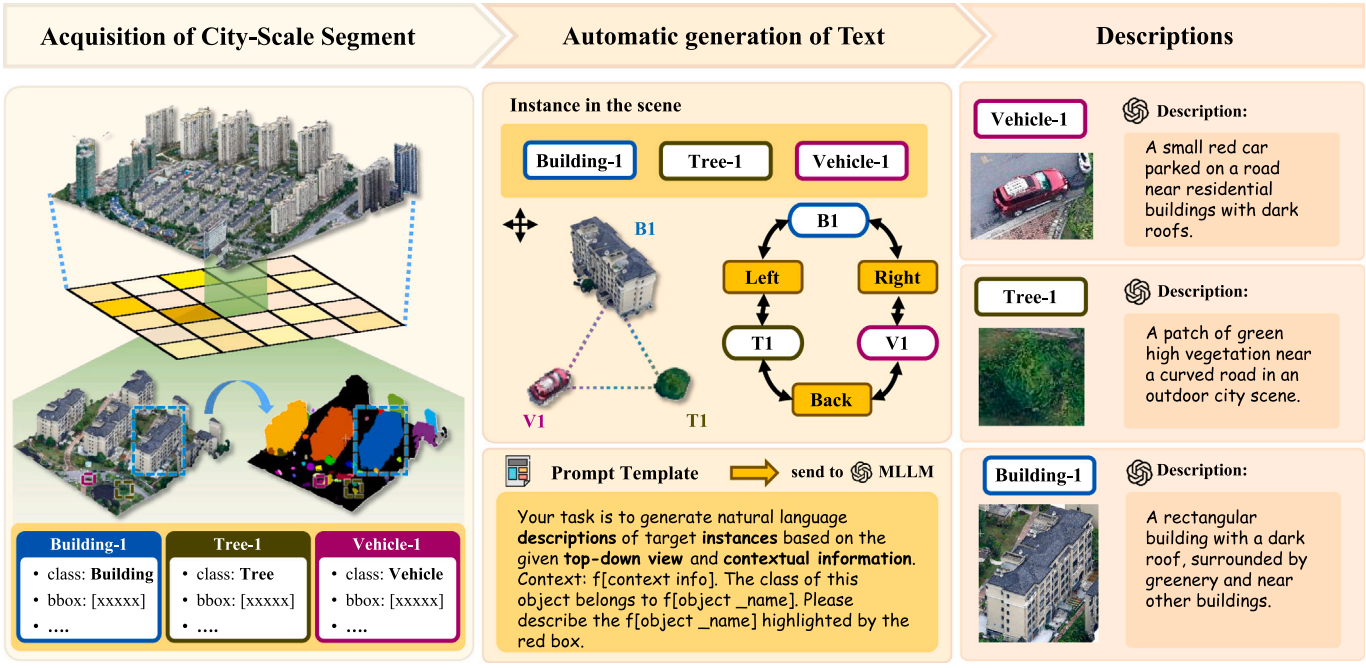


Fig. 2. Overview of the CityGrounding data construction process. The left side shows our spatial partitioning strategy. We perform instance segmentation on raw point clouds, compute inter-instance angular relations to build a scene graph, and use an MLLM to generate natural language descriptions from instance top-down views.

five representative urban categories, including buildings, vegetation, vehicles, trucks, and light poles.

In the second stage, we leverage Doubao-1.5-Vision-Pro to automatically generate textual descriptions for each instance based on top-down visual projections. Unlike previous LLM-assisted datasets that rely on handcrafted templates or manual post-correction, our annotation process is fully automated and guided by structured prompts designed to incorporate spatial relations, object semantics, and contextual reasoning. Specifically, we construct scene graphs that encode inter-object relationships, such as front, left, right, and back. We then embed this relational structure into the prompt to guide the MLLMs in producing spatially grounded and semantically rich descriptions. This results in textual annotations that not only describe visual appearance but also capture contextual and relational cues, which are essential for robust visual grounding.

To ensure the reliability of the annotations, we design an automatic filtering pipeline to remove ambiguous or low-confidence outputs, such as missing bounding boxes or irrelevant descriptions. The final annotations are stored in a ScanRefer-style JSON format to ensure compatibility with existing visual grounding pipelines.

We incorporate a prompt-based design paradigm into the dataset construction pipeline to ensure both quality and diversity of the generated descriptions. Motivated by the framework introduced in [36], we design a series of structured prompts that explicitly guide the MLLMs during generation. The prompt templates are presented in Table 2. The first instruction functions as a filtering stage, designed to remove instances that are invalid, ambiguous, or non-compliant with the task specifications following the generation process. Instructions 2, 3, 4, 7, 8, and 9 focus on controlling the type, syntactic structure, and length of the produced text, thereby ensuring grammatical correctness and consistency with the dataset's annotation style. Instructions 5 and 6, on the other hand, are designed to enhance semantic diversity and linguistic richness, encouraging the model to describe the same visual instance from multiple perspectives or to use varied expressions. Furthermore, we introduce an auxiliary variable `context_info`, which is provided as scene graph input to the multimodal large language model. This variable encodes contextual, spatial, and category-level information for each instance, enabling the model to generate more coherent and scene-aware textual descriptions. In this setup, `object_name` denotes the category label associated with the target instance, ensuring that the generated text remains semantically aligned with the corresponding visual content.

3.2. Dataset visualizations and statistics

In Fig. 3, we present an overview of our scene-level CityGrounding dataset. The dataset is constructed from high-resolution urban point clouds captured in two different cities, providing diverse large-scale outdoor environments. From these city-wide point clouds, we randomly selected several representative 100 m × 100 m blocks and performed instance-level segmentation to obtain object categories and 3D instance annotations. Subsequently, each instance is paired with a natural language description generated by a multimodal large language model, allowing us to build rich instance–text pairs that accurately reflect real urban spatial contexts.

We compare CityGrounding with existing benchmarks, including the indoor dataset ScanRefer [3] and the outdoor datasets KITTI360Pose [19] (a roadside dataset), CityRefer [29], and CityAnchor [22]. As shown in Table 1, our dataset achieves the highest point cloud density, making it the most precise outdoor urban-scale visual grounding dataset to date. In addition, the larger number of object instances extracted per unit area enables a finer and more comprehensive understanding of urban environments. In total, CityGrounding contains 32.9k natural language descriptions and 11,000 object instances.

Table 1
Comparison of existing 3D visual grounding datasets. L_{desc} indicates the average length of natural language descriptions, and $points/m^2$ represents the number of points per square meter in the scene.

Dataset	Scene	Source	Scale	Descs.	Objects	L_{desc}	pts/m^2
ScanRefer [3]	indoor	Scannet	Rooms	51.6k	11,046	20.3	–
Sr3D [2]	indoor	Scannet	Rooms	83.6k	8863	–	–
Nr3D [2]	indoor	Scannet	Rooms	41.6k	5879	11.4	–
KITTI360Pose [19]	outdoor	KITTI360	Roads.	43.3k	6800	7.6	73
CityQA [36]	outdoor	UrbanBIS	City	450k	–	–	–
NuScenes-QA [32]	outdoor	NuScenes	Roads.	460k	–	–	–
CityRefer [29]	outdoor	SensatUrban	City	35.2k	5866	26.3	474
CityAnchor [22]	outdoor	STPLS3D	City	1.4k	–	–	–
CityGrounding	**outdoor**	**City3D**	**City**	**32.9k**	**11,000**	**14.5**	**1110**

Table 2
The prompts used in dataset construction.

Task	Prompt
Dataset Construction	You are an expert in 3D visual grounding and natural language generation. Your core ability is to extract precise information from top-down views of point cloud data, identify targets marked by bounding boxes, and generate concise and accurate textual descriptions. Your task is to generate natural language descriptions of target instances based on the given top-down view and contextual information. You must follow these rules strictly: 1. If there is no bounding box, output only "no box" with no extra words. 2. If a bounding box exists, generate a clear and concise description reflecting the instance's key features. 3. Assume that the object is in an outdoor city scene. 4. The description should contain 20–30 words, with a natural and fluent style. 5. Include the entity's main attributes such as shape (e.g., rectangular, L shaped), color (e.g., reddish-brown roof), and identifiable structural details (e.g., "a small rectangular building with a slanted roof"). 6. Refer to nearby context if available (e.g., "near a Y shaped intersection", "behind another building"). 7. The output text should describe the object directly, not start with phrases like "This image …" or "Aerial view of …". 8. Descriptions for different instances should vary, avoid repetitive or template-like outputs. 9. Be strictly objective in describing the highlighted object. Do not include explanations or meta commentary. Context: `{context_info}`. The class of this object belongs to `{object_name}`. Please describe the `{object_name}` highlighted by the red box. Please generate 4 different natural language descriptions of this instance. Return them as a numbered list (1., 2., 3., 4.).

4. Method

In this section, we present the proposed framework in detail. As illustrated in Fig. 4, the overall architecture consists of two main components. First, the Graph-Structured Relation Extraction Module (GSREM) models inter-entity dependencies within the text to generate relation-aware embeddings, which are further refined through fusion with BiGRU-encoded features. For the 3D point cloud, the Neighbor-Guided Relation Enhancement (NGRE) module represents the scene with candidate objects, neighboring objects, and landmarks, each encoded by a SparseConv [10] backbone. The visual and textual features are then fused to obtain a unified multimodal representation, which is subsequently processed by a BiGRU [6] and a softmax layer to identify the target instance with the highest grounding score.

4.1. Neighbor-guided relation enhancement

In urban-scale 3D scenes, natural language descriptions inherently rely on rich spatial and semantic relations to uniquely identify target objects among visually similar distractors, for example, "the red car beside the tall building" or "the tree directly behind the delivery truck".

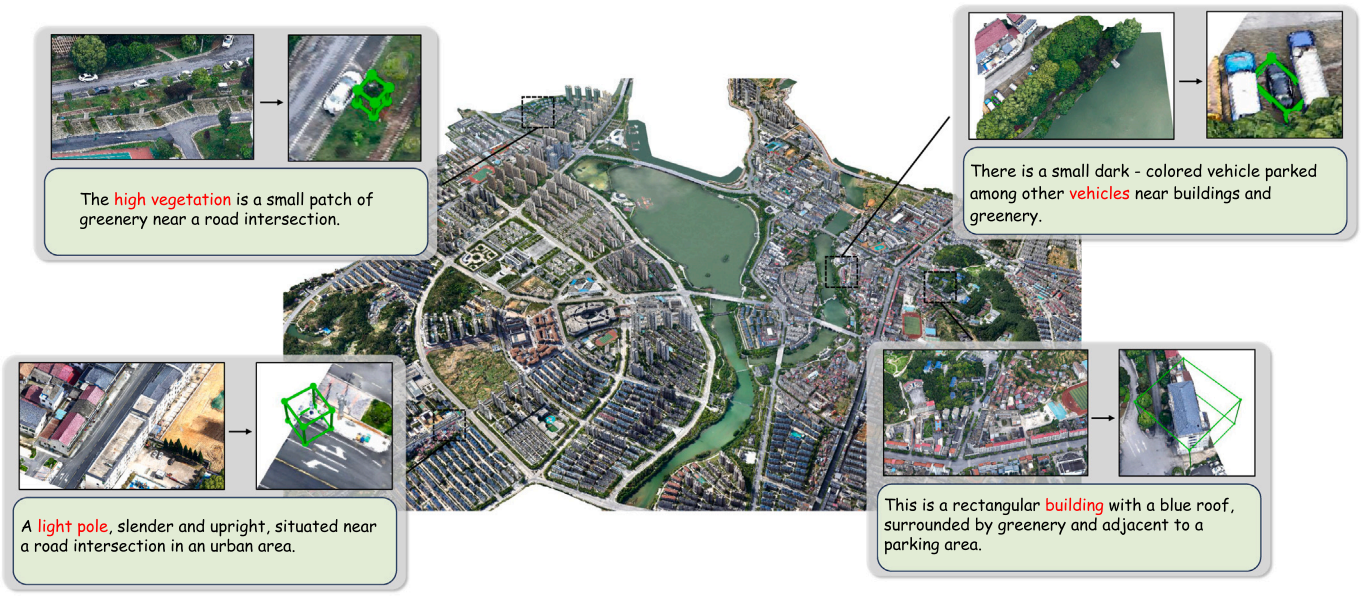


Fig. 3. Scene-level qualitative visualization of our dataset, shown in a representative urban environment.

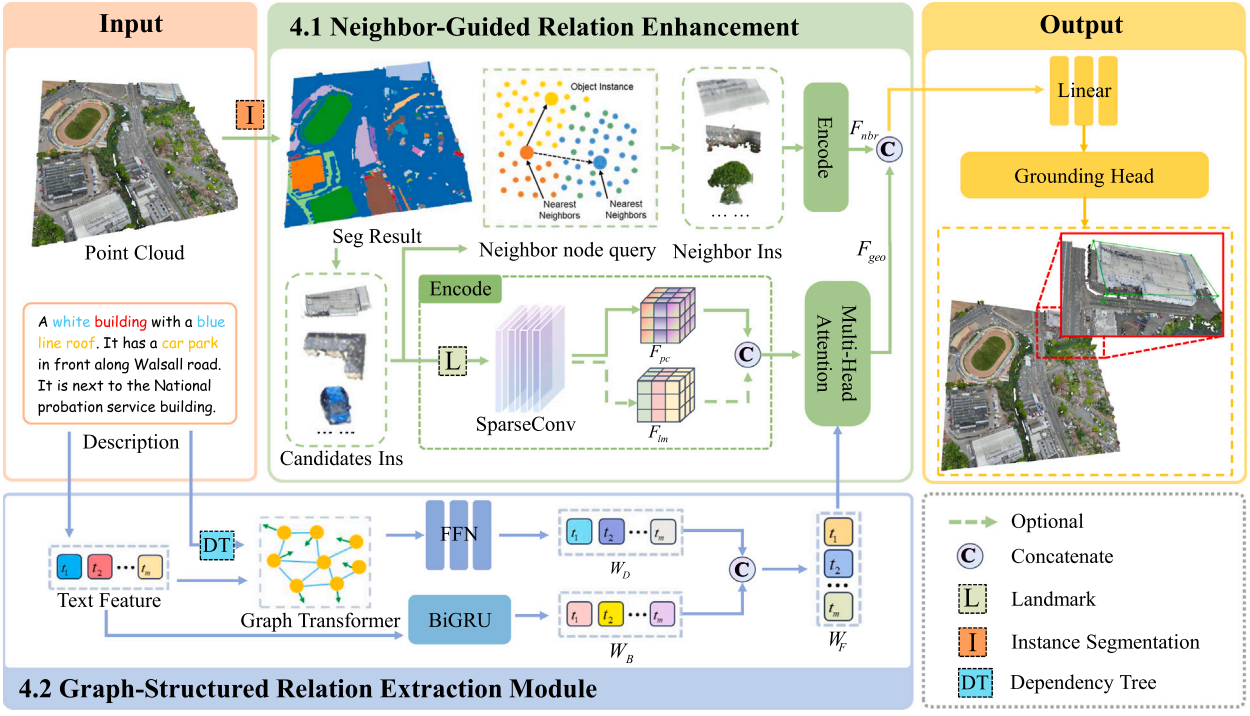


Fig. 4. Overview of the proposed framework. The model first extracts text embeddings using the Graph-Structured Relation Extraction Module. For the point cloud, candidate objects, neighboring objects, and landmarks are encoded by a SparseConv backbone. Landmark features are further enriched with their textual names. The fused features from text, candidate, neighboring, and landmark branches are processed by a BiGRU and softmax layer to predict the target object corresponding to the input text query.

In high-density urban environments, target objects are frequently surrounded by numerous visually and geometrically similar distractors. The traditional paradigm of treating objects as isolated entities is therefore insufficient.

To address this limitation, we propose a Neighbor-Guided Relation Enhancement (NGRE) module. NGRE shifts the learning objective from isolated feature extraction to neighborhood-aware geometric representation learning. Operating at the instance level rather than at the raw point level, it dynamically identifies the k nearest semantic entities (including candidate instances and critical landmarks) to construct a local reference system. Their features are encoded, and aggregated, and subsequently fused with the central candidate representation.

Specifically, to counter the severe visual ambiguity caused by dense distractors in urban scenes, we shift the paradigm from isolated object encoding to relational anchor construction. For each candidate object proposal o_b in the 3D scene, we dynamically establish a localized

reference system by identifying its k nearest semantic neighbors, denoted as $\mathcal{N}_K(o_b) = \{n_{b,k}\}_{k=1}^{K}$, where K is empirically set to 5.

For each neighboring instance $n_{b,k}$, we first voxelize its point cloud and represent it as a sparse tensor $S_{b,k}^{\mathrm{pc}} = (\mathbf{C}_{b,k}^{\mathrm{pc}}, \mathbf{X}_{b,k}^{\mathrm{pc}})$, where $\mathbf{C}_{b,k}^{\mathrm{pc}} \in \mathbb{Z}^{M_{b,k}\times 3}$ denotes the quantized voxel coordinates and $\mathbf{X}_{b,k}^{\mathrm{pc}} \in \mathbb{R}^{M_{b,k}\times d_{\mathrm{in}}}$ denotes the corresponding input point features. We then encode these localized visual anchors using SparseConv:

$$\hat{\mathbf{F}}_{\mathrm{pc}} = \mathrm{Pool}\left(\mathrm{SparseConv}\left(S_{b,k}^{\mathrm{pc}}\right)\right) \tag{1}$$

Similarly, for each neighboring landmark $\ell_{b,k}$, we construct its sparse tensor $S_{b,k}^{\mathrm{lm}} = (\mathbf{C}_{b,k}^{\mathrm{lm}}, \mathbf{X}_{b,k}^{\mathrm{lm}})$ and obtain:

$$\hat{\mathbf{F}}_{\mathrm{lm}} = \mathrm{Pool}\left(\mathrm{SparseConv}\left(S_{b,k}^{\mathrm{lm}}\right)\right) \tag{2}$$

where $\hat{\mathbf{F}}_{\mathrm{pc}} \in \mathbb{R}^{B\times K\times d}$ and $\hat{\mathbf{F}}_{\mathrm{lm}} \in \mathbb{R}^{B\times K\times d}$ denote the neighbor features of candidate objects and landmarks, respectively, with K being the number of neighbors and d the feature dimension. This deliberate instance-level aggregation explicitly captures relational geometric patterns and contextual semantics, anchoring the target candidate within its relevant surroundings. We then concatenate and aggregate these features to form a unified neighbor representation:

$$\mathbf{F}_{\mathrm{nbr}} = \mathrm{Fuse}([\hat{\mathbf{F}}_{\mathrm{pc}}, \hat{\mathbf{F}}_{\mathrm{lm}}]) \in \mathbb{R}^{B\times d} \tag{3}$$

where $\mathrm{Fuse}(\cdot)$ denotes mean pooling followed by a learnable transformation layer. This representation encodes the contextual relations surrounding each instance. Finally, to integrate this aggregated relational context with the inherent properties of the target instance, we perform a dimension-aligned feature fusion. Specifically, we expand $\mathbf{F}_{\mathrm{nbr}}$ to align with the feature dimensions of the target proposal. It is subsequently concatenated along the channel dimension with the object geometric features $\mathbf{F}_{\mathrm{geo}}$. The updated instance representation is formulated as follows:

$$\tilde{\mathbf{F}}_{\mathrm{geo}} = \mathrm{FuseContext}([\mathbf{F}_{\mathrm{geo}}, \mathbf{F}_{\mathrm{nbr}}]) \tag{4}$$

where $\tilde{\mathbf{F}}_{\mathrm{geo}}$ is the enhanced feature embedding that integrates both local instance features and neighbor-guided relational knowledge. This design allows the network to align with relational cues in textual descriptions, improving robustness when grounding objects described by relative positions or contextual dependencies.

4.2. Graph-structured relation extraction module

Existing visual grounding methods typically process text as a flat sequence. However, language queries in urban-scale scenes often contain deeply nested and multi-level spatial clauses, such as “behind the red car near the mall.” Relying solely on implicit semantic encoding can therefore lead to structural ambiguity and spatial misalignment. To address this problem, we introduce the Graph-Structured Relation Extraction Module (GSREM). We leverage a dependency parser as an Explicit Syntactic Topological Extractor. This intentional design choice provides a deterministic structural prior. Furthermore, we introduce an Edge-aware Attention mechanism that explicitly embeds the topological structure into the cross-modal attention matrix. This enables the network to perform logical deconstruction and reason over complex, multi-level relations (e.g., core subjects, multiple attributes, and spatial clauses).

Specifically, to effectively counteract the structural ambiguity inherent in processing complex urban queries as flat token sequences, we introduce a deterministic syntactic parsing stage. Given a natural language description of the target instance consisting of N sentences, rather than relying on implicit black-box semantic encoding, we strategically employ Stanford CoreNLP [28] as an Explicit Syntactic Topological Extractor. The parsed results are organized into a directed semantic graph representation, where word tokens serve as nodes and their precise grammatical dependencies define the relational edges. This explicit formulation is crucial for large-scale cityscapes, as it rigidly preserves multi-level syntactic dependencies and spatial cues that are frequently diluted by unconstrained global attention. Building upon this explicit structural prior, we employ an Edge-aware Graph Transformer Layer [8]. This module guides contextual information to propagate only along validated dependency pathways. This structurally constrained routing enables the extraction of logically disambiguated and relation-aware language features. The initial node and edge embeddings are constructed such that the textual features $\hat{t}_i$ are aligned with their corresponding indices to obtain the node representations $\hat{h}_i^0$. For the edge features β_{ij}, each dependency relation is assigned a unique identifier, which is further incorporated into the transformer and denoted as $\hat{e}_{ij}^0$, in order to facilitate relation-aware representation learning. The formula is:

$$\hat{h}_i^0 = \hat{t}_i, \quad \hat{e}_{ij}^0 = \beta_{ij}\mathbf{B}^0 + \mathbf{b}^0 \tag{5}$$

where $\mathbf{B}^0$ and $\mathbf{b}^0$ are the parameters of the linear projection layers. To capture relation-aware interactions, we apply a self-attention mechanism in which edge embeddings modulate the attention weights:

$$\alpha_{ij} = \mathrm{softmax}_j\left(\frac{(\hat{h}_i\mathbf{Q})(\hat{h}_j\mathbf{K})^{\top}}{\sqrt{d}} + \hat{e}_{ij}\mathbf{E}\right) \tag{6}$$

$$\mathrm{W}_D = \sum_{j\in N(i)} \alpha_{ij}(\hat{h}_j\mathbf{V}) \tag{7}$$

Next, the textual input is encoded using BERT to obtain token-level embeddings, which are subsequently projected into word representations via a linear layer. These representations are then processed by a BiGRU to extract features of the natural language description, denoted as W_B. Subsequently, these features are fused with the features W_D obtained from the Graph Transformer Layer, resulting in the enhanced feature representation W_F.

We modify the SWA (Superpoint–Word Aggregation) module [45] to enhance the alignment between textual and visual representations. In our adaptation, the attention weights can be modulated by an optional mask or edge embeddings, enabling the model to capture relational cues among objects in complex outdoor environments. Different from the original SWA, which relies on explicit padding masks, our version removes this component and instead focuses on edge-aware contextual aggregation, while retaining residual connections, layer normalization, and dropout to ensure stable and efficient learning. It is formulated as:

$$A = \mathrm{Softmax}\left(\frac{QK^{\top}}{\sqrt{d}} + E\right) \tag{8}$$

$$\mathrm{F}_D = \mathrm{LayerNorm}(Q + \mathrm{Dropout}(AV)) \tag{9}$$

where Q denotes the encoded candidate instance features, and K and V correspond to the encoded textual features.

4.3. Optimization objective

During training, we employ the Reference Loss to supervise the matching between predicted candidate instances and the ground-truth for each scene. This can be formulated as a Softmax Ranking Loss:

$$L_{\mathrm{ref}} = \frac{1}{B}\sum_{i=1}^{B} \mathrm{SoftmaxRankingLoss}(\mathbf{s}_i, \mathbf{y}_i) \tag{10}$$

where $\mathbf{s}_i$ denotes the predicted scores of candidate instances in scene i, and $\mathbf{y}_i$ is the corresponding ground-truth label.

5. Experiments

5.1. Experiments setting

To demonstrate the effectiveness of our design, we evaluate our approach on two established outdoor visual grounding benchmarks,

CityRefer and CityAnchor, as well as on our newly constructed dataset, CityGrounding. **CityRefer [29]** is built by augmenting the SensatUrban point cloud data with natural language descriptions for object instances. It covers $6km^2$ and includes 35k descriptions of 3D objects and 5k landmark annotations aligned with OpenStreetMap. The dataset contains four main categories: Building, Car, Ground, and Parking. Following the official protocol, we use the provided training and validation splits and report results on the validation set since the test set is unavailable. **CityAnchor [22]** is built by annotating 25 urban scene point clouds from STPLS3D with textual descriptions, resulting in 1448 instance–text pairs across nine categories, including Building, Vegetation, Vehicle, and LightPole. Following the official JSON release, we adopt the provided training split. For validation, the NO (Novel Objects) and ND (Novel Descriptions) subsets are merged, where NO includes unseen objects and ND contains familiar objects with novel descriptions. For the proposed **CityGrounding** dataset, we adopt a scene-level 8:2 split. Specifically, 367 scenes are used for training and 92 scenes are used as a held-out validation set. All descriptions and target instances from the same scene are assigned to the same split to avoid scene-level data leakage. All compared methods on CityGrounding are evaluated under this same split.

Our method is implemented in PyTorch. We use the Adam optimizer with a learning rate of 1×10^{-4} and a weight decay of $1e-5$. We evaluate the prediction accuracy by computing the Intersection over Union (IoU) between the predicted and ground-truth bounding boxes, with the IoU threshold set to 0.25 in all experiments.

5.2. Quantitative comparisons

Performance on CityRefer. As demonstrated in Table 3, we benchmark the performance of our proposed framework on the CityRefer dataset. Our approach yields a substantial improvement over the baseline, with the overall grounding accuracy reaching 58.29%, marking an absolute gain of 31.24%. Crucially, the most pronounced performance lift is observed in the Building and Car categories, registering improvements of 34.06% and 33.68%, respectively. This targeted effectiveness is significant, as Buildings and Cars represent critical and structurally complex entities in urban 3D visual grounding, demanding relational understanding.

Performance on CityGrounding. As indicated in Table 4, we evaluated our constructed dataset using both the CityRefer method and our proposed approach. Compared with the CityRefer method, our approach achieves a 1.93% improvement, reaching an overall accuracy of 69.83%.

Performance on CityAnchor. Because the CityAnchor dataset is designed for open-vocabulary 3D visual grounding, we merged the Novel Objects (NO) and Novel Descriptions (ND) subsets in the validation split to form a more comprehensive evaluation set. Under this setup, we conducted comparative experiments with CityRefer and our proposed CityRel-Net framework under identical configurations to ensure a fair comparison. As shown in Table 5, our method achieves a 3.87% improvement on the CityAnchor dataset, increasing the overall grounding accuracy from 47.81% to 51.68% compared to the previous SOTA.

These experimental results further validate the effectiveness of our proposed framework. Our method consistently improves grounding accuracy across multiple datasets by capturing dependency relations within textual descriptions and exploiting the relational context among neighboring instances. This capability enhances the alignment between language expressions and target instances, effectively capturing essential urban semantics and inter-object dependencies, thereby achieving state-of-the-art performance under the corresponding benchmark protocol and the best results among currently reported methods across the evaluated urban 3D visual grounding datasets.

Table 3
Evaluation (%) of 3D visual grounding methods on CityRefer. **Random** denotes random guessing results. **Humans** denote the accuracy achieved by human annotators based on textual descriptions. * denotes our reproduced results. For CityAnchor, EDA, and 3DVG-Transformer, we adopt the accuracy (%) reported in the CityAnchor paper on the CityRefer dataset, as it involves a modified version of the dataset.

Method	Building	Car	Ground	Parking	Overall
Random	10.30	10.30	9.10	9.40	10.10
CityRefer [29]	30.79	17.13	47.44	80.16	27.05
3DVG-Transformer [49]	–	–	–	–	7.73
EDA [46]	–	–	–	–	6.96
CityAnchor [22]	–	–	–	–	50.69
CityRel-Net (Ours)	**64.85**	**50.81**	**65.57**	**90.08**	**58.29**
InstanceRefer [47]	–	–	–	–	4.09
InstanceRefer* [47]	2.28	1.18	6.23	15.08	2.60
Humans	94.70	95.60	93.70	94.50	95.00

Table 4
Evaluation (%) of urban 3D visual grounding methods on CityGrounding. **Random** represents the results of the Uniform Random baseline, repeated 50 times, and reporting the mean accuracy (%).

Method	Years	Building	High Veg.	Light pole	Truck	Vehicle	Overall
Random	–	3.61	1.81	1.13	1.62	1.55	1.95
CityRefer [29]	2023	74.62	58.88	73.29	81.09	66.61	67.90
InstanceRefer [47]	2021	12.05	7.83	26.41	24.04	18.37	12.53
3D-SPS [26]	2022	8.95	0.63	4.45	3.53	1.59	4.34
CityRel-Net (Ours)	2026	**75.76**	**61.25**	**74.78**	**83.01**	**70.49**	**69.83**

Table 5
Performance comparison (%) on the CityAnchor dataset. As in our evaluation on CityRefer, the results of InstanceRefer, 3DVG-Transformer, EDA, and CityAnchor are reported according to the official accuracy (%) provided in the CityAnchor paper.

Method	Year	Venue	Overall
CityRefer [29]	2023	NeurIPS	6.07
InstanceRefer [47]	2021	ICCV	3.04
3DVG-Transformer [49]	2021	ICCV	6.25
EDA [46]	2023	CVPR	7.14
CityAnchor [22]	2025	ICLR	47.81
CityRel-Net (Ours)	2026	–	**51.68**

5.3. Ablation study

To validate the effectiveness of our framework, we conduct ablation studies on the Graph-Structured Relation Extraction Module (GSREM) and the Neighbor-Guided Relation Enhancement (NGRE) module, as summarized in Table 6. When both components are removed (row a), the model achieves only 27.05% accuracy, indicating that without explicit relational modeling, the baseline struggles to capture complex spatial and contextual dependencies in natural language descriptions. Introducing GSREM alone (row b) improves accuracy to 31.78%, confirming that GSREM effectively enhances the understanding of relations within textual descriptions by parsing linguistic structures into dependency graphs and modeling both inter-word and spatial relations within

Table 6
Performance gains (%) from ablating the GSREM and NGRE modules.

ID	GSREM	NGRE	Overall
(a)			27.05
(b)	✓		31.78
(c)		✓	50.37
(d)	✓	✓	**58.29**

Table 7
Ablation results (%) of varying the number of neighboring instances k in the NGRE module.

Number	Building	Car	Ground	Parking	Overall
3	64.07	49.47	**66.12**	88.10	57.38
5	**64.85**	**50.81**	65.57	**90.08**	**58.29**
7	**64.85**	43.74	65.38	86.11	54.38

textual descriptions. Using only NGRE (row c) improves the overall accuracy from 27.05% to 50.37%, corresponding to an absolute gain of 23.32 percentage points. This improvement reflects the joint effect of localized neighbor selection, geometric encoding, candidate and landmark context, and target-context fusion. Finally, the full model integrating both GSREM and NGRE (row d) achieves the highest overall accuracy of 58.29%, surpassing all ablated variants. These results demonstrate that the two modules are mutually complementary and jointly enable robust relational reasoning in large-scale urban visual grounding tasks.

To investigate the effect of the number of neighboring instances in the NGRE module, we conduct an ablation study by varying the neighbor count k. As shown in Table 7, increasing k from 3 to 5 improves grounding accuracy across multiple categories, indicating that a moderate number of neighboring instances helps capture more reliable contextual relations. However, when k is further increased to 7, the performance declines, particularly in the Car category. This is mainly due to the high density of objects, such as cars, in urban scenes, where incorporating too many neighboring instances introduces irrelevant or noisy contextual information. Overall, $k = 5$ achieves the best balance between relational diversity and noise robustness, leading to the highest overall accuracy (58.29%). Therefore, we adopt $k = 5$ as the default configuration in all experiments.

5.4. Qualitative results

Qualitative Analysis. We present qualitative comparisons between our method and the baseline CityRefer across the CityGrounding, CityRefer, and CityAnchor datasets in Figs. 5–7, respectively. Across



Fig. 5. Qualitative comparison between CityRefer and our CityRel-Net on the CityGrounding dataset.

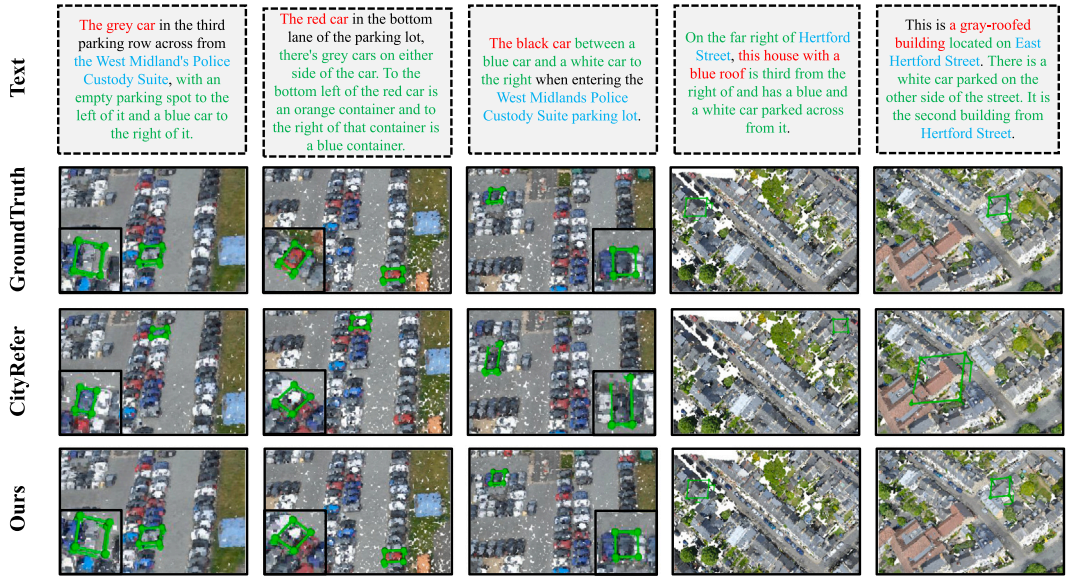


Fig. 6. Qualitative comparison between CityRefer and our CityRel-Net on the CityRefer dataset.

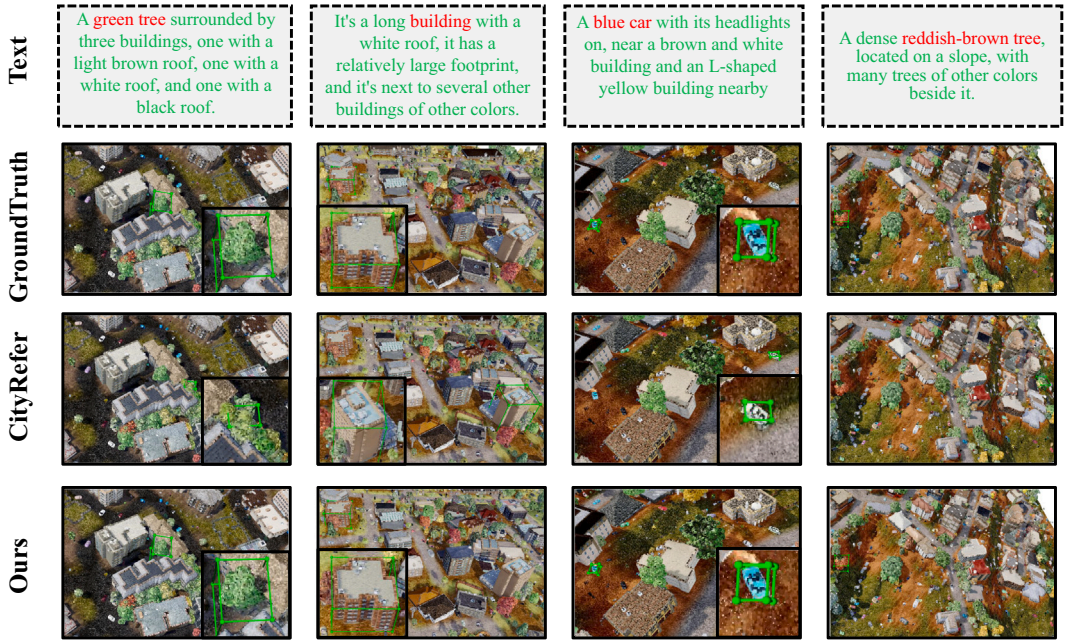


Fig. 7. Qualitative comparison between CityRefer and our CityRel-Net on the CityAnchor dataset.

these diverse benchmarks, our framework consistently demonstrates superior localization accuracy and stability.

Unlike baseline models that often misinterpret relational expressions or confuse closely spaced objects, our approach leverages relation-aware reasoning and neighbor-guided contextual cues to effectively disambiguate complex spatial dependencies, such as “behind”, “beside”, and “near”. This capability is particularly critical in dense urban scenes involving occlusion or clutter, where our method successfully distinguishes true referents from visually or semantically similar surrounding instances. Consequently, we achieve precise grounding for targets of varying scales ranging from large-scale structures (e.g., buildings, vegetation) to small, densely distributed entities (e.g., vehicles, street furniture). These consistent improvements across all three datasets validate the robustness of our framework in handling varying scene densities and category distributions.

6. Conclusion

In this paper, we present **CityRel-Net**, a relation-aware framework for urban-scale 3D visual grounding. To further evaluate grounding models in realistic dense urban environments, we construct CityGrounding, a real-world urban visual grounding dataset, featuring over 32.9 K language descriptions and 11 K object instances automatically annotated via multimodal large language models. Moreover, we introduce a relation-aware reasoning framework that integrates the Graph-Structured Relation Extraction Module (GSREM) and Neighbor-Guided Relation Enhancement (NGRE). GSREM explicitly models grammatical and spatial dependencies within textual descriptions, while NGRE captures inter-object relational contexts within dense point clouds. The proposed method enables fine-grained text–instance alignment and context-aware spatial reasoning, which are crucial for robust grounding in large-scale city scenes. Extensive experiments on CityRefer, CityAnchor, and the additional CityGrounding benchmark demonstrate that our approach achieves consistent state-of-the-art performance, with a remarkable +31.24% accuracy gain on CityRefer and superior robustness across diverse urban conditions. *We will continue to expand and refine the dataset to foster broader research and contribute more substantially to the community.*

CRediT authorship contribution statement

Xiaorong Zhang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Lingfei Chen:** Writing – review & editing, Visualization, Validation. **Yang Luo:** Writing – review & editing, Visualization. **Ting Han:** Writing – review & editing, Visualization. **Zilong Huang:** Writing – original draft, Visualization. **Guorong Cai:** Supervision, Resources, Funding acquisition. **Yiping Chen:** Supervision, Funding acquisition. **Jinhe Su:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported in part by the Natural Science Foundation of Xiamen, China, under Grant 3502Z202373036; in part by the National Natural Science Foundation of China under Grant 42371457; in part by the Natural Science Foundation of Fujian Province, China, under Grant 2025J01345.

Data availability

Data will be made available on request.

References

[1] A. Abdelreheem, U. Upadhyay, I. Skorokhodov, R. Al Yahya, J. Chen, M. Elhoseiny, 3dreftransformer: fine-grained object identification in real-world scenes using natural language, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2022, pp. 3941–3950.

[2] P. Achlioptas, A. Abdelreheem, F. Xia, M. Elhoseiny, L. Guibas, Referit3d: neural listeners for fine-grained 3D object identification in real-world scenes, in: European Conference on Computer Vision, Springer, 2020, pp. 422–440.

[3] D.Z. Chen, A.X. Chang, M. Nießner, Scanrefer: 3D object localization in RGB-D scans using natural language, in: European Conference on Computer Vision, Springer, 2020, pp. 202–221.

[4] M. Chen, Q. Hu, Z. Yu, H. Thomas, A. Feng, Y. Hou, K. McCullough, F. Ren, L. Soibelman, Stpls3d: a large-scale synthetic and real aerial photogrammetry 3D point cloud dataset, arXiv preprint arXiv:2203.09065, 2022.
[5] S. Chen, J. Fang, Q. Zhang, W. Liu, X. Wang, Hierarchical aggregation for 3D instance segmentation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 15467–15476.
[6] J. Chung, C. Gulcehre, K. Cho, Y. Bengio, Empirical evaluation of gated recurrent neural networks on sequence modeling, arXiv preprint arXiv:1412.3555, 2014.
[7] J. Devlin, M.W. Chang, K. Lee, K. Toutanova, BERT: pre-training of deep bidirectional transformers for language understanding, in: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers), 2019, pp. 4171–4186.
[8] V.P. Dwivedi, X. Bresson, A generalization of transformer networks to graphs, arXiv preprint arXiv:2012.09699, 2020.
[9] M. Feng, Z. Li, Q. Li, L. Zhang, X. Zhang, G. Zhu, H. Zhang, Y. Wang, A. Mian, Free-form description guided 3D visual graph network for object grounding in point cloud, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 3722–3731.
[10] B. Graham, M. Engelcke, L. Van Der Maaten, 3D semantic segmentation with submanifold sparse convolutional networks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 9224–9232.
[11] K. Guo, Y. Huang, T. Jia, X. Song, S. Sun, H. Wei, X.F. Han, S. Huang, N. Strisciuglio, S. Li, Visual grounding in 2D and 3D: a unified perspective and survey, Inf. Fusion (2025) 103625.
[12] K. Guo, H. Wei, Y. Huang, X. Song, S. Sun, M. Feng, H. Song, J. Li, Y. Zhang, ArgusNet: understanding 3D scenes more like humans, Neurocomputing (2026) 132895.
[13] P. Guo, H. Zhu, H. Ye, T. Li, T. Chen, Revisiting 3D visual grounding with context-aware feature aggregation, Neurocomputing 601 (2024) 128195.
[14] W. Guo, X. Xu, Z. Wang, J. Feng, J. Zhou, J. Lu, Text-guided sparse voxel pruning for efficient 3D visual grounding, in: Proceedings of the Computer Vision and Pattern Recognition Conference, 2025, pp. 3666–3675.
[15] Z. Guo, Y. Tang, R. Zhang, D. Wang, Z. Wang, B. Zhao, X. Li, Viewrefer: grasp the multi-view knowledge for 3D visual grounding, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 15372–15383.
[16] T. Hackel, N. Savinov, L. Ladicky, J.D. Wegner, K. Schindler, M. Pollefeys, semantic3d.net: a new large-scale point cloud classification benchmark, ISPRS Ann. Photogramm. Remote Sens. Spat. Inf. Sci. IV-1/W1 (2017) 91–98.
[17] Q. Hu, B. Yang, S. Khalid, W. Xiao, N. Trigoni, A. Markham, Sensaturban: learning semantics from urban-scale photogrammetric point clouds, Int. J. Comput. Vis. 130 (2022) 316–343.
[18] P.H. Huang, H.H. Lee, H.T. Chen, T.L. Liu, Text-guided graph neural networks for referring 3D instance segmentation, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2021, pp. 1610–1618.
[19] M. Kolmet, Q. Zhou, A. Ošep, L. Leal-Taixé, Text2pos: text-to-point-cloud cross-modal localization, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 6687–6696.
[20] L.H. Lee, T. Braud, P.Y. Zhou, L. Wang, D. Xu, Z. Lin, A. Kumar, C. Bermejo, P. Hui, et al., All one needs to know about metaverse: a complete survey on technological singularity, virtual ecosystem, and research agenda, Found. Trends® Hum.-Comput. Interact. 18 (2024) 100–337.
[21] F. Li, H. Jin, B. Gao, L. Fan, L. Jiang, L. Zeng, Nugrounding: a multi-view 3D visual grounding framework in autonomous driving, arXiv preprint arXiv:2503.22436, 2025.
[22] J. Li, H. Wang, J. Chen, Y. Liu, Z. Dou, Y. Ma, S. Yang, Y. Li, W. Wang, Z. Dong, B. Yang, et al., CityAnchor: city-scale 3D visual grounding with multi-modality LLMs, in: The Thirteenth International Conference on Learning Representations, 2025.
[23] D. Liu, Y. Liu, W. Huang, W. Hu, A survey on text-guided 3-D visual grounding: elements, recent advances, and future directions, IEEE Trans. Neural Netw. Learn. Syst. (2025) 1–22.
[24] Y. Liu, B. Sun, Y. Wang, J. Yang, X. Wang, F.Y. Wang, Talk to parallel lidars: a human-lidar interaction method based on 3D visual grounding, in: European Conference on Computer Vision, Springer, 2024, pp. 305–321.
[25] Z. Liu, Z. Zhang, Y. Cao, H. Hu, X. Tong, Group-free 3D object detection via transformers, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 2949–2958.
[26] J. Luo, J. Fu, X. Kong, C. Gao, H. Ren, H. Shen, H. Xia, S. Liu, 3D-SPS: single-stage 3D visual grounding via referred point progressive selection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 16454–16463.
[27] A.M. Mane, D. Weerakoon, V. Subbaraju, S. Sen, S.E. Sarma, A. Misra, GES3ViG: incorporating pointing gestures into language-based 3D visual grounding for embodied reference understanding, in: Proceedings of the Computer Vision and Pattern Recognition Conference, 2025, pp. 9017–9026.
[28] C.D. Manning, M. Surdeanu, J. Bauer, J.R. Finkel, S. Bethard, D. McClosky, The Stanford CoreNLP natural language processing toolkit, in: Proceedings of 52nd Annual Meeting of the Association for Computational Linguistics: System Demonstrations, 2014, pp. 55–60.
[29] T. Miyanishi, F. Kitamori, S. Kurita, J. Lee, M. Kawanabe, N. Inoue, CityRefer: geography-aware 3D visual grounding dataset on city-scale point cloud data, in: Proceedings of the 37th International Conference on Neural Information Processing Systems, 2023, pp. 77758–77770.
[30] M. Najibi, J. Ji, Y. Zhou, C.R. Qi, X. Yan, S. Ettinger, D. Anguelov, Unsupervised 3D perception with 2D vision-language distillation for autonomous driving, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 8602–8612.
[31] Q. Peng, H. Zheng, G. Huang, ProxyTransformation: preshaping point cloud manifold with proxy attention for 3D visual grounding, in: Proceedings of the Computer Vision and Pattern Recognition Conference, 2025, pp. 24582–24592.
[32] T. Qian, J. Chen, L. Zhuo, Y. Jiao, Y.G. Jiang, Nuscenes-qa: a multi-modal visual question answering benchmark for autonomous driving scenario, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2024, pp. 4542–4550.
[33] Z. Qian, Y. Ma, Z. Lin, J. Ji, X. Zheng, X. Sun, R. Ji, Multi-branch collaborative learning network for 3D visual grounding, in: European Conference on Computer Vision, Springer, 2024, pp. 381–398.
[34] H. Shi, M. Hayat, J. Cai, Unpaired referring expression grounding via bidirectional cross-modal matching, Neurocomputing 518 (2023) 39–49.
[35] X. Shi, Z. Wu, S. Lee, Aware visual grounding in 3D scenes, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 14056–14065.
[36] P. Sun, Y. Song, X. Liu, X. Yang, Q. Wang, T. Li, Y. Yang, X. Chu, 3D question answering for city scene understanding, in: Proceedings of the 32nd ACM International Conference on Multimedia, 2024, pp. 2156–2165.
[37] W. Tan, N. Qin, L. Ma, Y. Li, J. Du, G. Cai, K. Yang, J. Li, Toronto-3D: a large-scale mobile LiDAR dataset for semantic segmentation of urban roadways, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, 2020, pp. 202–203.
[38] G. Tziafas, H. Kasaei, Few-shot visual grounding for natural human-robot interaction, in: 2021 IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC), IEEE, 2021, pp. 50–55.
[39] T. Vu, K. Kim, T.M. Luu, T. Nguyen, C.D. Yoo, Softgroup for 3D instance segmentation on point clouds, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 2708–2717.
[40] C. Wang, W. Feng, X. Li, G. Cheng, S. Lyu, B. Liu, L. Chen, Q. Zhao, OV-VG: a benchmark for open-vocabulary visual grounding, Neurocomputing 591 (2024) 127738.
[41] X. Wang, N. Zhao, Z. Han, D. Guo, X. Yang, Augrefer: advancing 3D visual grounding via cross-modal augmentation and spatial relation-based referring, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2025, pp. 8006–8014.
[42] Y. Wang, Y. Li, S. Wang, G^3-LQ: marrying hyperbolic alignment with explicit semantic-geometric modeling for 3D visual grounding, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 13917–13926.
[43] C. Wu, J. Ji, H. Wang, Y. Ma, Y. Huang, G. Luo, H. Fei, X. Sun, R. Ji, et al., Rg-San: rule-guided spatial awareness network for end-to-end 3D referring expression segmentation, Adv. Neural Inf. Process. Syst. 37 (2024) 110972–110999.
[44] C. Wu, Y. Liu, J. Ji, Y. Ma, H. Wang, G. Luo, H. Ding, X. Sun, R. Ji, 3D-gres: generalized 3D referring expression segmentation, in: Proceedings of the 32nd ACM International Conference on Multimedia, 2024, pp. 7852–7861.
[45] C. Wu, Y. Ma, Q. Chen, H. Wang, G. Luo, J. Ji, X. Sun, 3D-stmn: dependency-driven superpoint-text matching network for end-to-end 3D referring expression segmentation, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2024, pp. 5940–5948.
[46] Y. Wu, X. Cheng, R. Zhang, Z. Cheng, J. Zhang, EDA: explicit text-decoupling and dense alignment for 3D visual grounding, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 19231–19242.
[47] Z. Yuan, X. Yan, Y. Liao, R. Zhang, S. Wang, Z. Li, S. Cui, Instancerefer: cooperative holistic understanding for visual grounding on point clouds through instance multi-level contextual referring, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 1791–1800.
[48] Y. Zhan, Y. Yuan, Z. Xiong, Mono3dvg: 3D visual grounding in monocular images, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2024, pp. 6988–6996.
[49] L. Zhao, D. Cai, L. Sheng, D. Xu, 3dvg-transformer: relation modeling for visual grounding on point clouds, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 2928–2937.
[50] Z. Zhu, X. Ma, Y. Chen, Z. Deng, S. Huang, Q. Li, 3D-Vista: pre-trained transformer for 3D vision and text alignment, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 2911–2921.

Author biography



Xiaorong Zhang received the B.S. degree from Jimei University, Fujian, China, in 2024. He is currently pursuing the M.S. degree with the School of Computer Engineering, Jimei University, under the supervision of Assoc. Prof. Jinhe Su. His main research interests include point cloud semantic segmentation and 3D visual grounding.



Lingfei Chen was admitted to Jimei University, Fujian, China, in 2025. She is currently pursuing the M.S. degree in an School of Computer Engineering, Jimei University, under the supervision of Assoc. Prof. Jinhe Su. Her main research interests include 3D visual grounding and vision-and-language navigation in large-scale urban scenes.



Yang Luo is currently pursuing his Ph.D. at the School of Geospatial Engineering and Science, Sun Yat-sen University, Zhuhai, China. He has published several papers in top journals and conferences, including IEEE T-GRS, IEEE T-ITS, IJDE, CVPR and AAAI. His research interests include point cloud semantic segmentation, 3D reconstruction, 4DGS, and its applications in remote sensing.



Ting Han (Graduate Student Member, IEEE) is currently pursuing his Ph.D. at the School of Geospatial Engineering and Science, Sun Yat-sen University, Zhuhai, China. He has published several papers in top journals and conferences, including IEEE T-IP, IEEE T-GRS, IEEE T-ITS, ISPRS Journal, JAG, CVPR, ICCV, and NeurIPS. His research interests include image and point cloud processing, semantic segmentation, 3D reconstruction, deep learning theory for 3D vision, and their applications in remote sensing and GeoAI.



Zilong Huang is currently pursuing a postgraduate degree at the School of Geospatial Engineering and Science, Sun Yat-sen University, Zhuhai, Guangdong, China. He has authored papers published in leading AI and computer vision conferences, including the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), the International Conference on Learning Representations (ICLR), and the Annual Conference on Neural Information Processing Systems (NeurIPS). His research interests include 3D generation, 3D reconstruction, and agentic systems for 3D scene generation.



Guorong Cai (Senior Member, IEEE) received the Ph.D. degree from Xiamen University, Xiamen, Fujian, China. He is currently a Professor with the School of Computer Engineering, Jimei University, Xiamen. His main research interests include computer vision and point cloud processing, including 3D reconstruction, image and point cloud-based object detection and recognition, stereo vision, deep learning theory, and its applications.



Yiping Chen (Senior Member, IEEE) received her Ph.D. degree in information and communication engineering from the National University of Defense Technology, Changsha, China, in 2011. From 2007 to 2011, she was an Assistant Researcher with The Chinese University of Hong Kong, Hong Kong. She is currently an Associate Professor with the School of Geospatial Engineering and Science, Sun Yat-Sen University, Zhuhai, China. She has co-authored more than 80 publications in IEEE Transactions on Geoscience and Remote Sensing, IEEE Transactions on Intelligent Transportation Systems, ISPRS Journal of Photogrammetry and Remote Sensing, and International Journal of Applied Earth Observation and Geoinformation, and in flagship Computer Vision (CV), and Artificial Intelligence conferences such as CVPR, AAAI and IJCAI. Her main research interests include image processing, mobile LiDAR data analysis, and GeoAI. Dr. Chen is an Associate Editor of IEEE Geoscience and Remote Sensing Magazine, IEEE Transactions on Geoscience and Remote Sensing and the Chair of ISPRS WG I/4 on LiDAR, Laser Altimetry, and Sensor Integration (2022–2026).



Jinhe Su received the Ph.D. degree from the University of Chinese Academy of Sciences, Beijing, China. He is currently an Associate Professor with the School of Computer Engineering, Jimei University, Xiamen, Fujian, China. His main research interests include machine learning, 3D reconstruction and remote sensing image processing.